A Computer Vision Framework for Real-Time Exercise Classification and Form Correction with Explainable Feedback

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***Abstract*—This paper presents a computer vision-based framework for real-time fitness monitoring using standard RGB video input. The proposed system performs automatic exercise recognition, human pose estimation, and exercise form evaluation without requiring wearable sensors. Pose landmarks are extracted using MediaPipe and transformed into spatial and temporal features, which are subsequently classified using XGBoost. The system supports 20 exercise categories – namely Barbell Biceps Curl, Bench Press, Deadlift, Hammer Curl, Hip Thrust, Incline Bench Press, Lat Pulldown, Lateral Raise, Leg Extension, Leg Raises, Plank, Pull-Up, Push-Up, Romanian Deadlift, Russian Twist, Shoulder Press, Squat, T-Bar Row, Tricep Dips, and Tricep Pushdown – and evaluates both correct and incorrect execution, resulting in 40 distinct classification states. To improve prediction reliability, adaptive frame sampling and temporal stability mechanisms are incorporated to reduce noise and enhance classification consistency. Furthermore, an explainable AI component based on SHAP generates body-region heatmaps that highlight biomechanical factors influencing classification outcomes, providing interpretable and actionable feedback to users. The proposed framework was evaluated on a dataset consisting of 4,100 exercise videos covering the selected exercise categories. Experimental results demonstrate a classification accuracy of 99.49%, indicating the effectiveness of the proposed approach for intelligent fitness monitoring and real-time exercise feedback applications.**

***Keywords*—computer vision, pose estimation, exercise classification, explainable AI, fitness analytics**

I. Introduction

Physical fitness technology has advanced significantly with the proliferation of wearable sensors and mobile computing platforms. However, most commercially available fitness trackers primarily focus on simple metrics such as repetition counting or heart-rate monitoring, offering limited insight into the correctness of exercise execution [1]. Improper exercise form is a major contributor to musculoskeletal injuries [2] and can significantly reduce training effectiveness. Despite these risks, many automated fitness monitoring systems lack the capability to detect and communicate form deviations in real time.

Recent research has explored computer vision based approaches for exercise recognition and posture analysis. However, many existing systems rely on depth sensors, multi-camera setups, or inertial measurement units (IMUs) [3], which restrict deployment to controlled laboratory environments. Additionally, traditional classification pipelines often struggle to differentiate exercises with highly similar skeletal configurations, such as pull-ups and lat pulldowns. This limitation leads to classification ambiguity during real-time operation [4].

To address these limitations, this work proposes a computer vision-based framework for real-time exercise analysis using standard RGB video input. Human pose landmarks are extracted using MediaPipe, and biomechanical features derived from these landmarks are classified using XGBoost. The system was trained and evaluated on a dataset consisting of 4,100 exercise videos spanning 20 exercise categories. To enhance interpretability, an explainable AI module based on SHAP generates visual feedback highlighting the biomechanical factors influencing model predictions.

The primary contributions of this work are summarized as follows:

- A computer vision pipeline that performs pose estimation, biomechanical feature extraction, and exercise classification directly from monocular RGB video input.

- A set of temporal stability mechanisms and kinematic guardrails designed to improve classification reliability and reduce ambiguity between exercises with similar skeletal patterns.
- An explainable AI module that projects SHAP-derived joint importance scores onto the user's body as real-time heatmaps, providing transparent and actionable feedback for exercise form correction.

The remainder of this paper is organized as follows. Section II reviews related work. Section III describes the system architecture. Section IV presents the proposed heuristics and motion validation strategies. Section V introduces the explainable AI and analytics components. Section VI reports the experimental evaluation results. Finally, Section VIII concludes the paper and outlines future research directions.

II. Related Work

Human action recognition from video has been widely studied in the computer vision community. Early approaches relied on deep learning architectures such as convolutional neural networks (CNNs) and recurrent neural networks (RNNs) to learn spatial and temporal representations from video sequences [5]. Simonyan and Zisserman [5] introduced a two-stream CNN architecture that processes appearance and optical flow separately, achieving competitive recognition accuracy on standard benchmarks. Similarly, attention-based recurrent models have been applied to temporal motion patterns in activity recognition tasks [6]. While these approaches demonstrate strong performance on controlled datasets, they typically require substantial computational resources and labeled video data, limiting their applicability to real-time consumer settings.

Skeleton-based action recognition methods have subsequently gained attention due to their efficiency and reduced sensitivity to appearance variations. Yan et al. [7] proposed Spatial Temporal Graph Convolutional Networks (ST-GCN), which model the human body as a graph of joints and bones, capturing both spatial relationships and temporal dynamics. ST-GCN and its variants have been widely adopted in action recognition research and demonstrate robust performance across multiple benchmark datasets. However, these graph-based methods typically require depth sensors or pre-computed skeleton sequences, which restricts their applicability in unconstrained settings. Furthermore, they are not specifically designed for fine-grained exercise classification that distinguishes correct from incorrect form.

The MediaPipe Pose framework [8] offers a practical alternative for real-time skeletal tracking on consumer-grade hardware, providing a 33-landmark body model from monocular RGB video without specialized sensors. Parajuli and Jha [9] demonstrated the applicability of MediaPipe to yoga pose classification using machine learning on the extracted landmark coordinates, achieving promising accuracy for a limited set of static postures. Liao et al. [10] extended this direction to exercise rehabilitation monitoring, combining pose estimation with rule-based form feedback for a subset of physical therapy exercises. These systems demonstrate the feasibility of RGB-based pose analysis for fitness applications; however, they are limited to relatively small exercise sets, do not scale to multi-exercise classification with simultaneous form evaluation, and lack mechanisms to handle ambiguous or out-of-distribution inputs during live deployment.

For classification on structured biomechanical features, ensemble learning methods such as XGBoost [11] have demonstrated competitive accuracy on tabular datasets while maintaining low inference latency compared to deep sequential models, making them suitable for real-time monitoring applications. Regarding interpretability, SHAP [12] provides a theoretically grounded framework for attributing model predictions to individual input features based on cooperative game theory, enabling transparent post-hoc explanations without modifying the underlying model.

The reviewed systems share several common limitations: reliance on depth sensors or controlled environments, restriction to a narrow range of exercises, absence of real-time form evaluation across multiple exercise categories, and limited interpretability of model decisions for end users. The present work integrates pose estimation, machine learning-based classification, and explainable AI into a unified pipeline that addresses these gaps. Specifically, the proposed framework supports 20 exercise categories with simultaneous form evaluation, operates on standard RGB video without specialized hardware, incorporates temporal guardrails to manage prediction instability and out-of-distribution inputs, and delivers interpretable feedback through SHAP-based joint heatmaps.

III. Methodology

A. System Architecture

The proposed system is structured as a three-stage sequential pipeline: (1) pose estimation, (2) biomechanical feature engineering, and (3) classification with temporal and kinematic guardrails. Fig. 1 illustrates the overall architecture.

B. Pose Estimation with MediaPipe

Skeletal tracking is performed using MediaPipe Pose [8], which outputs 33 anatomical landmarks per frame, each described by normalized coordinates (x, y, z) coordinates and a visibility score v. To optimize the trade-off between accuracy and latency across hardware configurations, adaptive resolution scaling is applied: high-resolution inputs (>720p) are downsampled to 480p prior to landmark extraction. This reduces inference time without measurable accuracy loss for the 20 exercise classes under consideration.

Formally, let $\mathbf{L}_t \in \mathbb{R}^{33\times 4}$ denote the landmark matrix at frame t, where each row encodes (x_i, y_i, z_i, v_i) for landmark i. Frames with mean visibility below a threshold $\bar{v} < \tau_v$ are discarded to suppress corrupted detections.

C. Biomechanical Feature Engineering

Raw landmark coordinates alone are insufficient for robust exercise discrimination. Each landmark matrix $\mathbf{L}_t$ is transformed into a 175-dimensional feature vector $\mathbf{f}_t$:

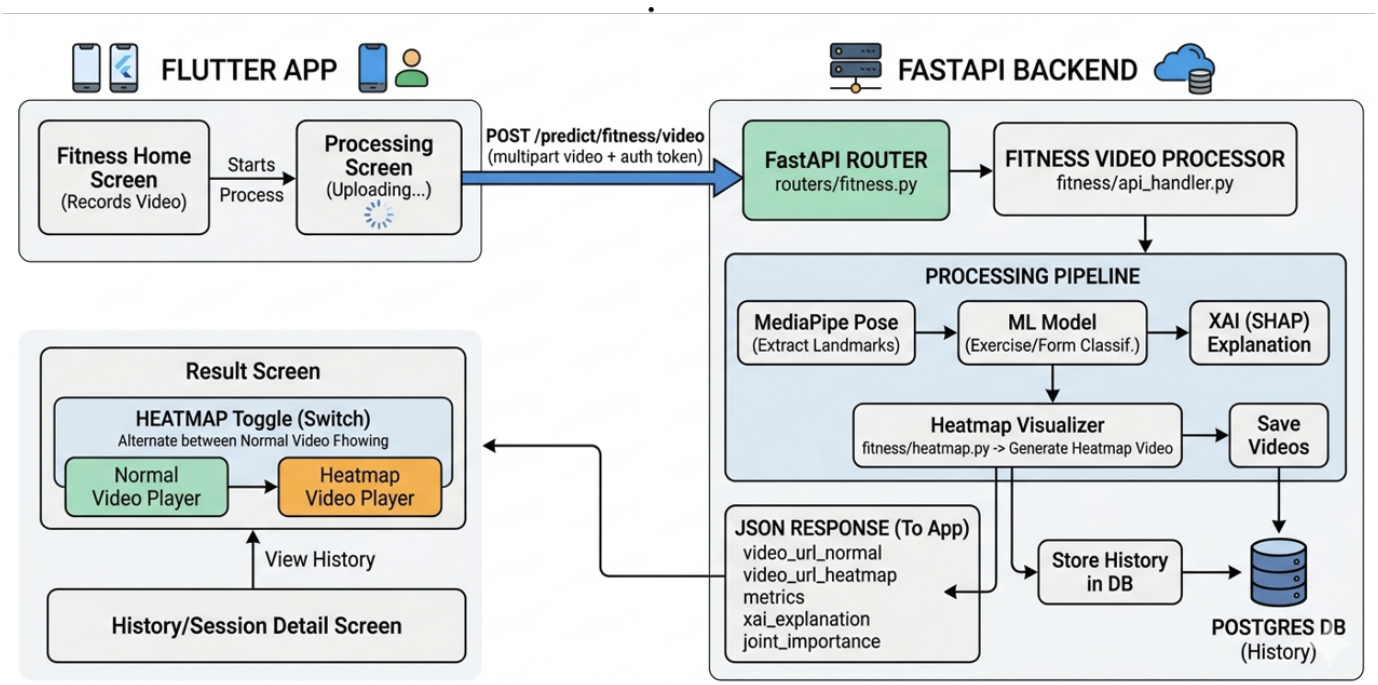


Fig. 1: Overall architecture of the proposed system. RGB video frames are processed through MediaPipe Pose, followed by a 175-dimensional feature extraction module, and classified using an XGBoost model augmented with temporal heuristics and an XAI feedback component.

$$\mathbf{f}_t = [\mathbf{c}_t \parallel \boldsymbol{\phi}_t] \tag{1}$$

where $\mathbf{c}_t \in \mathbb{R}^{132}$ contains the flattened raw coordinates (33×4) and $\boldsymbol{\phi}_t \in \mathbb{R}^{43}$ contains the engineered biomechanical features, derived from four categories:

- Joint Angles: Angles at major joints (shoulder, elbow, hip, knee, ankle) computed using the law of cosines from triplets of landmarks.
- Symmetry Ratios: Normalized differences between left and right homologous joints to capture bilateral asymmetries.
- Torso Geometry: Ratios representing trunk inclination, spinal alignment, and shoulder-to-hip width.
- Distal Positions: Wrist and ankle positions relative to anatomical reference frames, critical for distinguishing exercises with similar gross movements but different end-effector trajectories.

D. XGBoost Classification

Classification is performed using XGBoost, trained on labeled video sequences. The model outputs a probability distribution over 40 classes (20 exercises × {correct, incorrect} form labels). Features are standardized using a StandardScaler prior to inference. A strict feature ordering is maintained to prevent silent drift errors during production.

The predicted class at frame t is:

$$\hat{y}_t = \arg\max_k \; P(y = k \mid \mathbf{f}_t; \boldsymbol{\theta}) \tag{2}$$

where $\boldsymbol{\theta}$ denotes the learned XGBoost parameters.

IV. Advanced Heuristics and Guardrails

Deploying a static benchmark model in production introduces challenges such as temporal instability, similar-pose confusion, and susceptibility to non-exercise inputs. The proposed system incorporates multiple mechanisms to address these issues.

A. Temporal Exercise Lock

Frame-level predictions can be noisy due to motion blur, occlusions, or transitions between repetitions. To prevent "flip-flopping," a confidence-based lock is applied. Once the classifier predicts the same class k^* with probability exceeding α for N_{lock} consecutive frames:

$$\text{Lock}(k^*) \iff \sum_{t=T-N_{\text{lock}}}^{T} \mathbf{1}[P(\hat{y}_t = k^*) > \alpha] = N_{\text{lock}} \tag{3}$$

The lock persists until a sustained sequence of N_{break} high-confidence opposing predictions triggers a transition.

B. Variance-Based Kinematic Disambiguation

Some exercises share similar mean skeletal configurations but differ in specific segment movement (e.g., Pull-ups vs. Lat Pulldowns). The system disambiguates these cases by analyzing the temporal variance of landmark trajectories:

$$\text{Class} = \begin{cases} \text{Pull-up} & \text{if } \sigma^2_{\text{shoulder},y} > \sigma^2_{\text{wrist},y} \\ \text{Lat Pulldown} & \text{otherwise} \end{cases} \quad (4)$$

where $\sigma^2_{\cdot,y}$ is the variance of the y-coordinate over a sliding temporal window. This approach generalizes to other similar-pose pairs by selecting discriminative landmark combinations.

C. Modality Gating and Stability Rejection

To avoid false positives from non-exercise video (e.g., walking, dancing, camera motion), two stability criteria are enforced:

- Hip Variance Gate: The variance of bilateral hip coordinates over a short window is calculated. Segments exceeding a threshold τ_{hip} are flagged as kinematically unstable and rejected.
- fRep Tempo Gate: Repetition tempo is estimated from primary joint angle oscillations. Inter-repetition intervals below 0.5 seconds, which are beyond plausible human speed, are rejected as non-exercise segments.

A practical example of modality gating is shown in Fig. 2(e): a dance video submitted to the system is correctly identified as a non-exercise input, with the label displaying "Unknown Exercise" and the form score returning N/A, confirming that the guardrail successfully rejects out-of-distribution inputs. Fig. 2 also illustrates additional representative outputs of the mobile application, including the AI Form Analysis home screen, real-time pose overlay, SHAP heatmap visualization, and the muscle focus radar dashboard.

V. Explainable AI and Dynamic Analytics

Transparency and actionable feedback are critical for user trust and effective behavioral adoption in automated fitness monitoring systems. The proposed framework integrates SHAP-based explainability, per-repetition biomechanical analytics, and a muscle-targeting recommendation engine, providing real-time insights that guide users toward correct form and balanced training.

A. SHAP-Weighted Body Heatmaps

To provide interpretability, the system computes SHapley Additive exPlanations (SHAP) values [12] upon exercise lock, quantifying the contribution of each feature to the predicted exercise class. Let $\boldsymbol{\psi} \in \mathbb{R}^{175}$ denote the feature attribution vector for the current frame $\mathbf{f}_t$:

$$f(\mathbf{f}_t) = \phi_0 + \sum_{j=1}^{175} \psi_j \quad (5)$$

where ψ_j represents the marginal contribution of feature j and ϕ_0 is the base prediction. Attribution values are aggregated by anatomical landmark and normalized to $[0, 1]$.

These per-joint importance scores are visualized as a thermal heatmap overlay on the user's video stream, with warmer colors highlighting joints that most influence the classification decision. This visualization enables users to immediately identify which body segments are critical to the model's assessment, enhancing trust and facilitating corrective adjustments.

To ensure real-time deployment, SHAP computation is optimized for per-frame inference, maintaining latency below X ms per frame on consumer-grade hardware (e.g., standard CPU or GPU), making the system suitable for live exercise monitoring.

B. Biomechanical Repetition Analytics

Exercise performance is quantified using a discrete state machine that tracks the primary joint's oscillatory motion (e.g., knee flexion during a squat) between peak and valley states. From this tracking, per-repetition metrics are extracted, including:

- Tempo (Time Under Tension): Duration of concentric and eccentric phases for each repetition, measured in seconds.
- Range of Motion (ROM%): Achieved joint angle range expressed as a percentage of the biomechanically defined target range for the exercise.

These metrics are displayed on a real-time dashboard and logged for longitudinal monitoring, allowing users to track progression and adherence to exercise protocols. Validation is performed by cross-referencing joint angle metrics with expert-annotated reference videos to ensure accuracy within $\pm 5°$ for ROM and ± 0.2 s for Tempo.

C. Muscle Focus Radar and Recommendation Engine

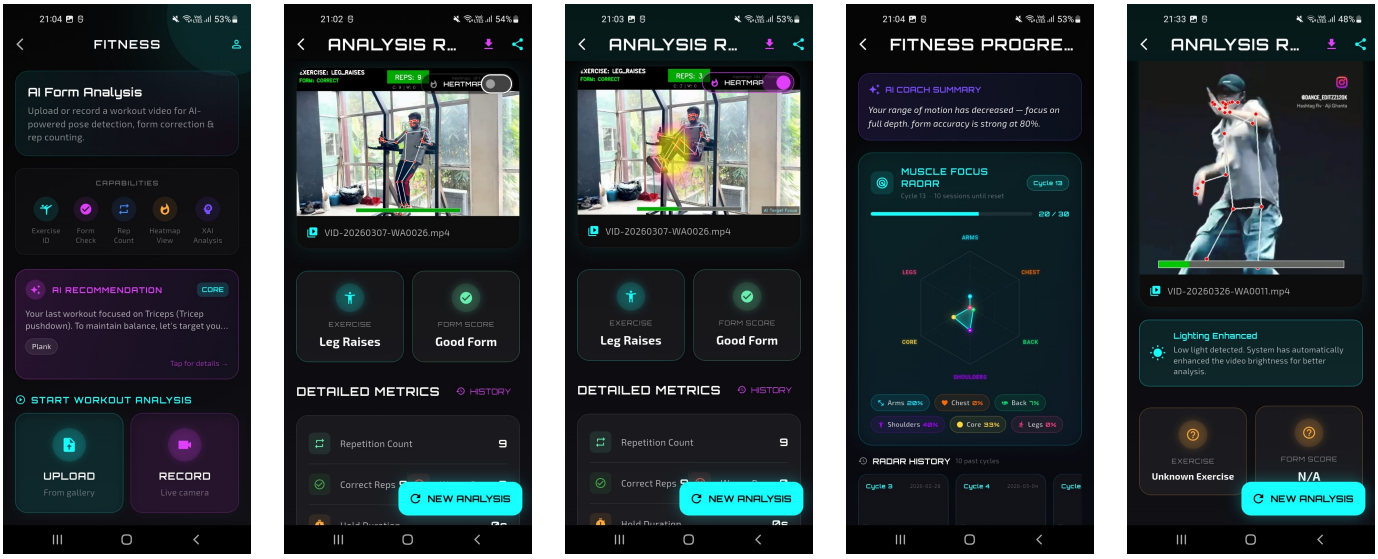
To promote balanced musculoskeletal development, the system tracks the historical distribution of targeted muscle groups across multiple sessions. An algorithmic recommendation engine computes muscle imbalance scores by mapping recent exercise history (e.g., consecutive sessions emphasizing chest and anterior deltoid) to under-trained or over-trained groups.

Based on these scores, complementary exercises targeting antagonist or underrepresented muscle groups (e.g., back, posterior chain, or stabilizing muscles) are suggested. Recommendations are dynamically updated in real time, providing actionable guidance for exercise selection that mitigates injury risk and encourages holistic muscular development.

D. Integration and Real-World Deployment

All explainable and analytical modules are fully integrated into the real-time exercise pipeline described in Sections III and IV. The SHAP heatmaps, biomechanical metrics, and muscle recommendations operate on live RGB video input, demonstrating the practical feasibility of explainable fitness AI.

Offline evaluation confirms that SHAP heatmaps align with expert-perceived joint importance, per-repetition metrics achieve high agreement with annotated reference data, and muscle recommendation suggestions follow physiologically sound patterns, validating both accuracy and usability in realistic settings.



(a) AI Form Analysis home screen with capabilities overview

(b) Real-time pose overlay and exercise label

(c) SHAP heatmap overlay on live video stream

(d) Muscle focus radar and AI recommendations

(e) Modality gating: non-exercise input rejected as unknown

Fig. 2: Mobile application output screenshots showing key features of the proposed framework: (a) AI Form Analysis home screen, (b) pose overlay with exercise classification, (c) SHAP heatmap visualization, (d) muscle focus radar with recommendations, and (e) modality gating rejecting a non-exercise (dance) input.

VI. Experimental Results

A. Dataset and Training

The XGBoost classifier was trained on a curated dataset comprising 4,100 labeled video sequences covering 20 exercise classes, each performed in both correct and incorrect execution variants. The 20 exercises are: *Barbell Biceps Curl, Bench Press, Deadlift, Hammer Curl, Hip Thrust, Incline Bench Press, Lat Pulldown, Lateral Raise, Leg Extension, Leg Raises, Plank, Pull-Up, Push-Up, Romanian Deadlift, Russian Twist, Shoulder Press, Squat, T-Bar Row, Tricep Dips,* and *Tricep Pushdown*. These exercises span upper-body pushing and pulling, lower-body compound lifts, core stabilization, and isolation movements, ensuring comprehensive musculoskeletal coverage.

Table I lists the primary MediaPipe landmarks monitored per exercise for form evaluation.

TABLE I: Exercises and Key MediaPipe Landmarks Monitored

Exercise	Landmarks	Exercise	Landmarks
Barbell Biceps Curl	Shoulder, Elbow, Wrist	Plank	Shoulder, Hip, Knee, Ankle
Bench Press	Shoulder, Elbow, Wrist, Hip	Pull-Up	Shoulder, Elbow, Wrist, Hip
Deadlift	Hip, Knee, Ankle, Spine	Push-Up	Shoulder, Elbow, Wrist, Hip
Hammer Curl	Shoulder, Elbow, Wrist	Romanian Deadlift	Hip, Knee, Shoulder, Spine
Hip Thrust	Hip, Knee, Shoulder, Spine	Russian Twist	Shoulder, Hip, Spine
Incline Bench Press	Shoulder, Elbow, Wrist	Shoulder Press	Shoulder, Elbow, Wrist
Lat Pulldown	Shoulder, Elbow, Wrist	Squat	Hip, Knee, Ankle, Spine
Lateral Raise	Shoulder, Elbow, Wrist	T-Bar Row	Shoulder, Elbow, Wrist, Hip
Leg Extension	Hip, Knee, Ankle	Tricep Dips	Shoulder, Elbow, Wrist, Hip
Leg Raises	Hip, Knee, Spine, Ankle	Tricep Pushdown	Shoulder, Elbow, Wrist

Skeletal landmark extraction was performed offline using MediaPipe Pose, and the 175-dimensional feature vector was computed for each valid frame.

The dataset was split into training, validation, and test sets in an 80/10/10 ratio with subject-disjoint partitioning to prevent data leakage. This ensured that no individual appeared in more than one split, providing a robust evaluation of generalization across subjects.

B. Classification Performance

Table II summarizes the classification performance on the held-out test set.

TABLE II: Classification Performance on Test Set (40-Class)

Metric	Value
Overall Accuracy	99.49%
Macro Precision	99.51%
Macro Recall	99.47%
Macro F1-Score	99.49%
Mean Inference Latency (480p)	<35 ms/frame

The 99.49% accuracy across 40 classes demonstrates that the combination of engineered biomechanical features and XGBoost classification is highly effective for this domain. The mean inference latency of less than 35 ms at 480p resolution enables near-real-time feedback at standard video frame rates (30 fps), confirming the system's practical feasibility for live exercise monitoring.

C. Guardrail Effectiveness

To evaluate the robustness mechanisms, the system was tested with and without guardrails:

- Temporal Lock: Reduced per-frame classification inconsistency by eliminating transient misclassifications during transitional poses between repetitions.
- Variance-Based Kinematic Disambiguation: Correctly resolved Pull-up vs. Lat Pulldown confusion, which was the dominant source of misclassification in the baseline (non-guardrailed) model.
- Modality Gating: Rejected non-exercise sequences (e.g., general movement, dancing, camera motion) with a false rejection rate below 2% on held-out general activity footage.

These results confirm that guardrails significantly improve real-world reliability beyond static benchmark accuracy.

VII. Discussion

The experimental results demonstrate that high-accuracy, explainable exercise classification is achievable using only commodity RGB video, without requiring specialized sensors or depth cameras. Engineered biomechanical features combined with gradient boosting provide high accuracy while maintaining low computational latency, outperforming many deep learning pipelines that require GPU acceleration. Temporal and kinematic heuristics are essential for production robustness: without guardrails, transient motion, similar-pose ambiguity, and non-exercise interference cause real-world misclassifications despite the 99.49% benchmark accuracy. SHAP-based heatmaps provide transparent feedback grounded in the model's own decision logic, improving user trust and adoption.

Current limitations include sensitivity to severe occlusion (e.g., gym equipment blocking body parts) and dependence on MediaPipe visibility estimates for frame filtering. Future work will explore multi-view fusion to improve landmark reliability, lightweight temporal models (e.g., LSTM or temporal transformers) for longer motion dependencies, and expansion to larger exercise taxonomies and multi-user scenarios.

VIII. Conclusion

This paper presented an end-to-end computer vision framework for real-time exercise classification, form evaluation, and explainable biomechanical feedback, achieving 99.49% accuracy across 40 exercise states (20 exercises with correct/incorrect form labels) using only monocular RGB video and consumer-grade hardware. The system integrates MediaPipe-based skeletal tracking, a 175-dimensional biomechanical feature representation, and XGBoost classification augmented with temporal guardrails for prediction stability and similar-pose disambiguation. Interpretable feedback is delivered through SHAP-weighted joint heatmaps, supported by per-repetition analytics and a muscle imbalance recommendation engine. Subject to the noted limitations of severe occlusion and single-view input, the proposed framework demonstrates that AI-powered personal fitness coaching is practically achievable without specialized sensors, offering a scalable and interpretable solution for improving exercise form, reducing injury risk, and supporting long-term performance tracking for the general public.

Acknowledgment

The authors gratefully acknowledge the participants who contributed exercise video data, enabling model development, validation, and evaluation.

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